

Controlling Pause Behavior When Fine-Tuning VITS-Based Text-to-Speech

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Abstract

Fine-tuning a VITS-based text-to-speech model on expressive read speech reproduces the speaker's pause habits, including over-long dramatic pauses. The stochastic duration predictor adds a second problem: pause tails vary by hundreds of milliseconds between identical renders. This paper describes three practical interventions developed while fine-tuning Piper on 1,546 clips of an expressive speaker (Espresso, speaker ex02). First, a measurement protocol: at noise_w=0.5 the median pause is stable but the tail is not, so every sentence is rendered three times and the median is reported. Second, pause-normalized fine-tuning: internal silences longer than 300 ms in the training audio are shortened to 200 ms before training; after 2 h 18 min of continued training, comma pauses fell from p50 330 ms to 140 ms (teacher: 180 ms) and the general-distribution tail collapsed from 960 ms to 280 ms. A control run — 2.69 h of continued training on the unmodified data — left pauses near the stage-1 level, attributing the collapse to the normalization rather than to more training. Third, an opt-in synthesis-time pause clamp that hard-bounds intra-piece pauses without retraining. Voice quality was preserved: 11 vs 12 word errors against the teacher, UTMOS 4.25 vs 4.35, and 0.74 similarity to the speaker's own recordings. Negative results are reported honestly: a learned pause model that only learned punctuation, and a long-form retrain that failed for lack of phrase-break signal.

1 Introduction

VITS models learn pause durations from their training data through a stochastic duration predictor. When the training speaker is a dramatic reader, the model reproduces dramatic pauses: comma pauses roughly twice as long as a neutral teacher voice, with tails approaching a full second. Separately, the predictor's randomness makes single-render pause measurements unreliable.

This paper documents the interventions used to bring pauses under control while fine-tuning Piper, a VITS-based model, on Espresso speaker ex02: a measurement protocol, a data-side fix, and an inference-time guardrail. All three are small, cheap, and independent of the fine-tuning framework. Two abandoned approaches are reported as negative results.

2 Setup

The student is a Piper medium voice (about 15.8M parameters, 63.5 MB as ONNX), fine-tuned from en_US-amy-medium. The data is the Espresso dataset (CC BY-NC 4.0), speaker ex02, 1,546 quality-gated read clips. Training ran with PyTorch Lightning on a Modal L4 GPU. Evaluation uses 40 targeted sentences (20 comma /

10 semicolon / 10 colon) plus 10 general sentences; pauses are measured as internal silences via energy-based detection; word errors via Whisper small.en; naturalness via UTMOS; voice similarity via ECAPA-TDNN speaker embeddings. Unless noted, inference uses noise_scale=0.3, noise_w=0.5.

3 The stochasticity problem

At the default noise_w=0.8, rendering the same sentence twice produced different pause placements: phantom pauses appearing and disappearing between renders. Reducing the duration-noise knob to noise_w=0.5 (with noise_scale=0.3) removed the phantom pauses with no retraining, an inference-only change. But the predictor stays stochastic: two identical runs of the pause evaluation gave different tails for the same voice (Table 1).

run	comma p50	comma p95	comma max	general max
1	270 ms	420 ms	420 ms	500 ms
2	280 ms	529 ms	700 ms	980 ms

Table 1: Two identical pause evaluations of the stage-1 voice. The median is stable; the tail swings by hundreds of milliseconds.

The median is stable (270 to 280 ms) but the tail swings by hundreds of milliseconds run to run. At noise_w=0.5, a single render's longest pause is therefore not a reliable measurement. The protocol adopted for everything below: render each sentence three times and take the median for targeted sentences (pooling for the general distribution), which removes the tail noise.

4 Pause-normalized fine-tuning

Internal silences longer than 300 ms in the 1,546 gated clips were shortened to 200 ms with an audio-domain edit; the text was left unchanged. The distribution shift is shown in Table 2: the median pause is untouched (60 ms), while the tail is cut down (p95 420 to 207 ms, max 1,820 to 300 ms).

split	n pauses	p50	p95	max
before	6,521	60 ms	420 ms	1,820 ms
after	6,534	60 ms	207 ms	300 ms

Table 2: Pause distribution in the training clips before and after normalization.

Stage 2 fine-tunes the stage-1 checkpoint on this normalized set (2 h 18 min of new GPU time, in two legs; comma pauses had already visibly collapsed after 0.65 h). Results are in Table 3 and Figure 1. Stage 1 had learned comma pauses roughly twice the teacher's (p50 330 vs 180 ms; p95 443 vs 260 ms) and produced general-distribution pauses up to about a second. After stage 2, the comma distribution sits inside the teacher's range (p50 140 vs 180 ms; p95 205 vs 260 ms; max 300 vs 260 ms), and the general-distribution tail collapsed (p95 260 vs 600 ms; max 280 vs 960 ms). The stock libritts_r voice, which barely pauses at commas at all (p50 45 ms), is included as a lower reference.

voice	comma p50	comma p95	comma max	general p50	general p95	general max
amy (teacher)	180	260	260	140	220	340
ex02 stage 1	330	443	500	200	600	960
ex02 stage 2	140	205	300	140	260	280
libritts_r 3922	45	80	80	120	310	400

Table 3: Pause distributions (ms) across voices. All rows use the 3-repeat protocol.

One measurement caveat: the targeted punctuation rows use the longest internal silence per sentence as a proxy for the mark's pause, which can select an unrelated pause. The general-distribution rows (all internal silences of at least 100 ms) have no such proxy problem and tell the same story.

Ablation: continued training without normalization

To isolate the normalization from the effect of additional training, the stage-1 checkpoint was trained for a further 2.69 h on the unmodified dataset with identical hyperparameters (two legs; an L4 preemption was resumed from the run's own checkpoint). The resulting voice sits near stage 1 on every tail metric and far from stage 2 (Table 4). A small convergence effect is visible (comma p50 330 to 270 ms), but it does not explain the collapse into the teacher's range. The improvement is therefore attributed to the pause normalization.

voice	comma p50	comma p95	comma max	general p95	general max
stage 1	330	443	500	600	960
stage 2 (normalized, +2.31 h)	140	205	300	260	280
ablation (unmodified, +2.69 h)	270	448	600	506	680

Table 4: Ablation: continued training on the unmodified data leaves pauses near stage-1 levels (ms).

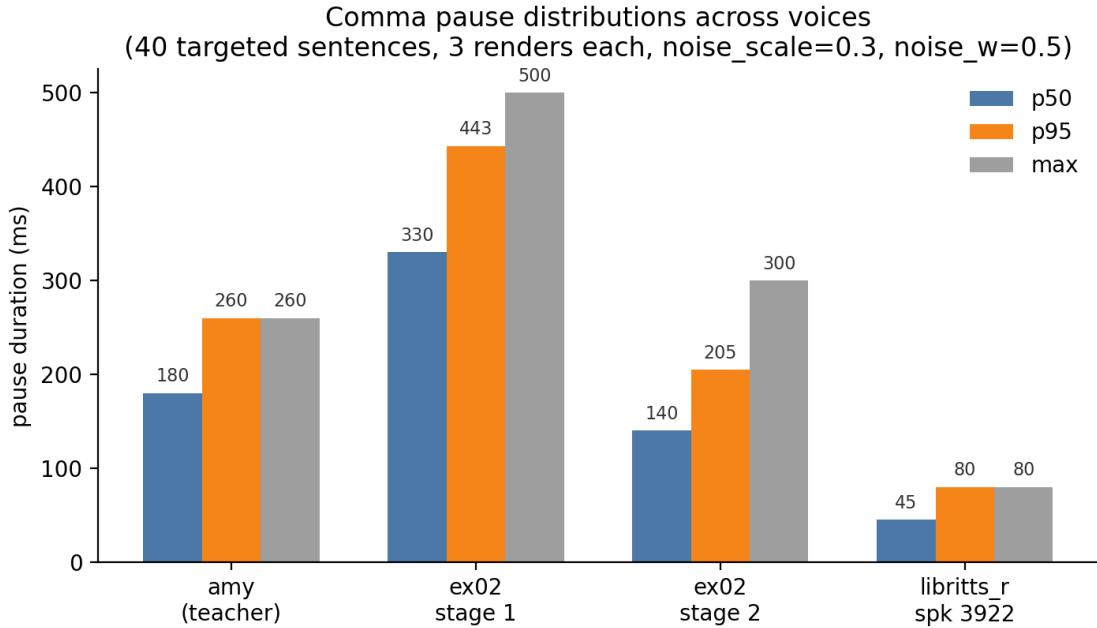


Figure 1: Comma pause distributions across voices (Table 3, first three columns).

5 Voice quality is preserved

Pause surgery must not damage the voice. Table 5 reports the standard 40-sentence evaluation (30 held-out Espresso gate texts plus reference and word-test sentences) for the teacher and the stage-2 voice. All six pre-registered checks pass: no more word errors than the teacher, UTMOS within 0.3 of the teacher (breakage alarm only), one clean voice (0.74 similar to the speaker's own recordings, against her own consistency of 0.48; 0.30 similar to amy, against a 0.37 ceiling), and time-to-first-audio and throughput within 10% of the teacher.

metric	amy (teacher)	ex02 stage 2
word errors (lower better)	12	11
UTMOS naturalness	4.35	4.25
similarity to real ex02	0.25	0.74
similarity to amy	0.81	0.30
time to first audio	144 ms	107 ms
throughput	11.3x realtime	13.3x realtime

Table 5: Standard voice metrics, teacher vs stage-2 voice.

Against the stage-1 baseline (17 word errors, UTMOS 4.01), stage 2 is cleaner on both counts while keeping voice identity (0.74 vs 0.75 similarity to the speaker's recordings).

6 Synthesis-time pause clamp

As a guardrail independent of training, the pacing layer gained an opt-in `max_pause_ms`: internal silences longer than the cap are shortened by removing middle frames with a 3 ms crossfade. Leading and trailing silence and the hand-written inter-piece pauses are untouched; the clamp applies within synthesized pieces only. Table 6 shows the effect on the stage-1 voice at a 250 ms cap.

voice	comma p95	comma max	general p95	general max
stage 1, clamp off	443	500	600	960
stage 1, clamp on	260	260	260	280

Table 6: Pause clamp (250 ms cap) on the stage-1 voice, in ms.

The clamp is a hard upper bound on intra-piece pauses; it cannot fix pauses the model never produces (libritts_r's 45 ms commas are untouched), and inter-piece pauses set by the pacing table remain as written.

7 Negative results

A 93.6k-parameter BiLSTM pause model reached F1 0.934 on held-out data, but inspection showed it had mostly learned punctuation-to-pause mapping: 99.2% of its training pauses follow a comma or a period. It is kept as an opt-in; the hand-written table remains the default.

A long-form retrain on 5.56 hours of audiobook narration failed cleanly (F1 0.354): only 1.4% of pauses in that data were not after punctuation, so there was no phrase-break signal to learn. Long-form data also did not buy word correctness: the stock libritts_r-medium voice renders "semicolon" as "Sima", while the ex02 and amy voices render it correctly.

8 Limitations

This is a single-speaker, single-architecture case study, not a general claim. The ablation isolates the normalization from the convergence effect for this speaker, but other speakers and architectures may behave differently. Listening evaluation so far is informal; a blind A/B between the two finalist checkpoints is in progress, with listening as the final quality gate.

9 Conclusion

Three cheap interventions — a 3-repeat measurement protocol, pause-normalized fine-tuning, and an opt-in synthesis-time clamp — brought a dramatic reader's pauses into the teacher's range without harming word correctness, naturalness, or voice identity. A controlled ablation attributes the improvement to the normalization rather than to additional training. The failures are part of the result: in this regime, pause behavior is dominated by punctuation, and neither a learned model nor long-form data changed that with the data available.

References

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